C3.1 Integration of body systems

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/ahl-c3-1.html

Written for this site from the class slides, with textbook page references for each statement. The chapter is on pages 509–517 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Tropisms

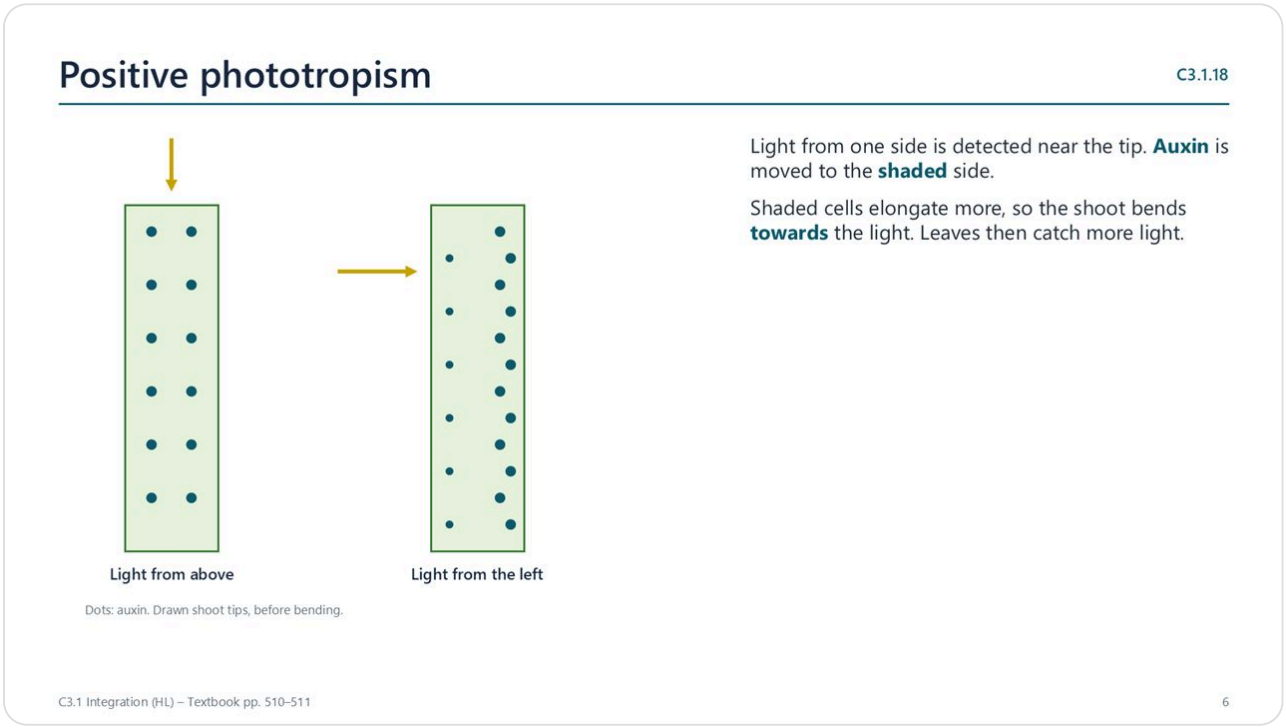
C3.1.17 Observing tropic responses

pp. 509–510

Grow seedlings in the dark, then light them from one side. Record drawings at intervals and measure the angle of curvature with a protractor. Use seedlings lit from above and seedlings kept in the dark as controls, and find the mean of many seedlings.

C3.1.18 Positive phototropism

pp. 510–511



Phototropins in the shoot tip detect light from one side. Auxin is moved to the shaded side, where cells elongate more, so the shoot bends towards the light.

Phytohormones

C3.1.19 Phytohormones

pp. 512–513

Plants use many signalling chemicals: auxin (elongation, apical dominance), cytokinin (cell division, side buds), gibberellin (stem growth, germination), ethylene (ripening), abscisic acid (stomatal closure) and jasmonic acid (defence).

C3.1.20 Auxin efflux carriers

pp. 514–515

Auxin diffuses into cells but cannot diffuse out. It leaves through **efflux carriers** placed on one side of each cell. When neighbouring cells use the same side, auxin is pumped through the tissue in one direction and builds up a concentration gradient.

C3.1.21 Auxin and cell growth

p. 515

Auxin makes cells pump H^+ into their walls. The acid loosens cross-links between cellulose molecules, so turgor stretches the cell. Unequal auxin causes unequal growth, which bends the shoot.

C3.1.22 Auxin and cytokinin

p. 516

Shoot tips make auxin, which moves to the roots; root tips make cytokinin, which moves to the shoots. Their interaction keeps root and shoot growth in proportion, for example in apical dominance.

C3.1.23 Ethylene and ripening

pp. 516–517

Ethylene stimulates ripening, and ripening fruit makes more ethylene: **positive feedback**. Ripening is fast and synchronized, so many fruits are ready together to attract seed dispersers.

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